

```
[1]: !pip install numpy
```

Requirement already satisfied: numpy in c:\users\admin\anaconda3\lib\site-packages (2.3.5)

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

```
[3]: df = pd.read_csv('diwali sales data.csv', encoding = 'unicode_escape')
```

```
[4]: df.shape
```

```
[4]: (11251, 15)
```

```
[5]: df.head(5)
```

File

```
[4]: df.shape
```

[4]: (11251, 15)

```
[5]: df.head(5)
```

[5]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Categ
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	/
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	/
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	/
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	/
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	/

```
[6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   User_ID               11251 non-null  int64
1   Cust_name             11251 non-null  object
2   Product_ID           11251 non-null  object
3   Gender               11251 non-null  object
4   Age Group            11251 non-null  object
5   Age                  11251 non-null  int64
6   Marital_Status       11251 non-null  int64
7   State                11251 non-null  object
8   Zone                 11251 non-null  object
9   Occupation           11251 non-null  object
10  Product_Category     11251 non-null  object
11  Orders               11251 non-null  int64
12  Amount               11239 non-null  float64
13  Status               0 non-null      float64
14  unnamed1             0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

```
[9]: df.drop(['Status', 'unnamed1'], axis=1, inplace = True)
```

```
[10]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 11251 entries, 0 to 11250
```

```
[9]: df.drop(['Status', 'unnamed1'], axis=1, inplace = True)
```

```
[10]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   User_ID               11251 non-null  int64
1   Cust_name             11251 non-null  object
2   Product_ID           11251 non-null  object
3   Gender                11251 non-null  object
4   Age Group             11251 non-null  object
5   Age                   11251 non-null  int64
6   Marital_Status        11251 non-null  int64
7   State                 11251 non-null  object
8   Zone                  11251 non-null  object
9   Occupation            11251 non-null  object
10  Product_Category      11251 non-null  object
11  Orders                11251 non-null  int64
12  Amount                11239 non-null  float64
dtypes: float64(1), int64(4), object(8)
memory usage: 1.1+ MB
```

```
[11]: pd.isnull(df).sum()
```

```
[11]: User_ID          0
      Cust_name       0
      Product_ID      0
      Gender          0
      Age Group       0
      Age             0
      Marital_Status  0
      State           0
      Zone            0
      Occupation      0
      Product_Category 0
      Orders          0
      Amount          12
      dtype: int64
```

```
[12]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   User_ID          11251 non-null  int64
1   Cust_name        11251 non-null  object
2   Product_ID       11251 non-null  object
3   Gender           11251 non-null  object
```

```
[13]: df.dropna(inplace = True)
```

```
[14]: pd.isnull(df).sum()
```

```
[14]: User_ID          0
      Cust_name      0
      Product_ID     0
      Gender         0
      Age Group      0
      Age            0
      Marital_Status 0
      State          0
      Zone           0
      Occupation     0
      Product_Category 0
      Orders         0
      Amount         0
      dtype: int64
```

```
[19]: attendance_list = [['AMIT',23], ['ANKIT',21], ['KESHAV', ], ['Tarun', 25]]

      df_attendance = pd.DataFrame(attendance_list, columns =['Name','Age'] )
```

```
[20]: df_attendance
```

```
[20]:
```

	Name	Age
0	AMIT	23.0
1	ANKIT	21.0
2	KESHAV	NaN
3	Tarun	25.0

```
[23]: df_attendance.dropna(inplace = True)
```

```
[24]: df_attendance
```

```
[24]:
```

	Name	Age
0	AMIT	23.0
1	ANKIT	21.0
3	Tarun	25.0

```
[25]: df['Amount'] = df['Amount'].astype('int')
```

```
[26]: df['Amount'].dtypes
```

```
[26]: dtype('int64')
```



```
[27]: df.columns
```

```
[27]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
        'Orders', 'Amount'],  
        dtype='object')
```

```
[34]: df.rename(columns = {'Marital_Status':'Shadi'},inplace = True)
```

```
[35]: df.columns
```

```
[35]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Shadi', 'State', 'Zone', 'Occupation', 'Product_Category', 'Orders',  
        'Amount'],  
        dtype='object')
```

```
[36]: df.rename(columns = {"Shadi":"Marital_Status"},inplace = True)
```

```
[37]: df.columns
```

```
[37]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
        'Orders', 'Amount'],  
        dtype='object')
```

```
[27]: df.columns
```

```
[27]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
        'Orders', 'Amount'],  
        dtype='object')
```

```
[34]: df.rename(columns = {'Marital_Status': 'Shadi'}, inplace = True)
```

```
[35]: df.columns
```

```
[35]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Shadi', 'State', 'Zone', 'Occupation', 'Product_Category', 'Orders',  
        'Amount'],  
        dtype='object')
```

```
[36]: df.rename(columns = {"Shadi": "Marital_Status"}, inplace = True)
```

```
[37]: df.columns
```

```
[37]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
        'Orders', 'Amount'],  
        dtype='object')
```

```
[38]: df[['Amount', 'Age', 'Orders']].describe()
```



```
[26]: df['Amount'].dtypes
```

```
[26]: dtype('int64')
```

```
[27]: df.columns
```

```
[26]: df['Amount'].dtypes
```

```
[26]: dtype('int64')
```

```
[27]: df.columns
```

```
[27]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
        'Orders', 'Amount'],  
        dtype='object')
```

```
[34]: df.rename(columns = {'Marital_Status':'Shadi'},inplace = True)
```

```
[35]: df.columns
```

```
[35]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Shadi', 'State', 'Zone', 'Occupation', 'Product_Category', 'Orders',  
        'Amount'],  
        dtype='object')
```

```
[25]: df[ 'Amount' ].dtypes
```

```
[26]: dtype('int64')
```

```
[27]: df.columns
```

```
[27]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
         'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
         'Orders', 'Amount'],  
        dtype='object')
```

```
[34]: df.rename(columns = {'Marital_Status':'Shadi'},inplace = True)
```

```
[35]: df.columns
```



```
[35]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
         'Shadi', 'State', 'Zone', 'Occupation', 'Product_Category', 'Orders',  
         'Amount'],  
        dtype='object')
```

```
[36]: df.rename(columns = {"Shadi":"Marital_Status"},inplace = True)
```

```
[37]: df.columns
```

```
[37]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
         'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
         'Orders', 'Amount'],
```

```
df = df[['Amount', 'Age', 'Orders']]
```

```
[38]: df[['Amount', 'Age', 'Orders']].describe()
```

```
[38]:
```

	Amount	Age	Orders
count	11239.000000	11239.000000	11239.000000
mean	9453.610553	35.410357	2.489634
std	5222.355168	12.753866	1.114967
min	188.000000	12.000000	1.000000
25%	5443.000000	27.000000	2.000000
50%	8109.000000	33.000000	2.000000
75%	12675.000000	43.000000	3.000000
max	23952.000000	92.000000	4.000000

```
[47]: count = 11
while count < 28:
    print(count)
    count += 5
```

```
11
```

```
16
```

```
[68]: butter_price = {'amul': 30, 'mother dairy' : 24 , 'urmul': 29}
```

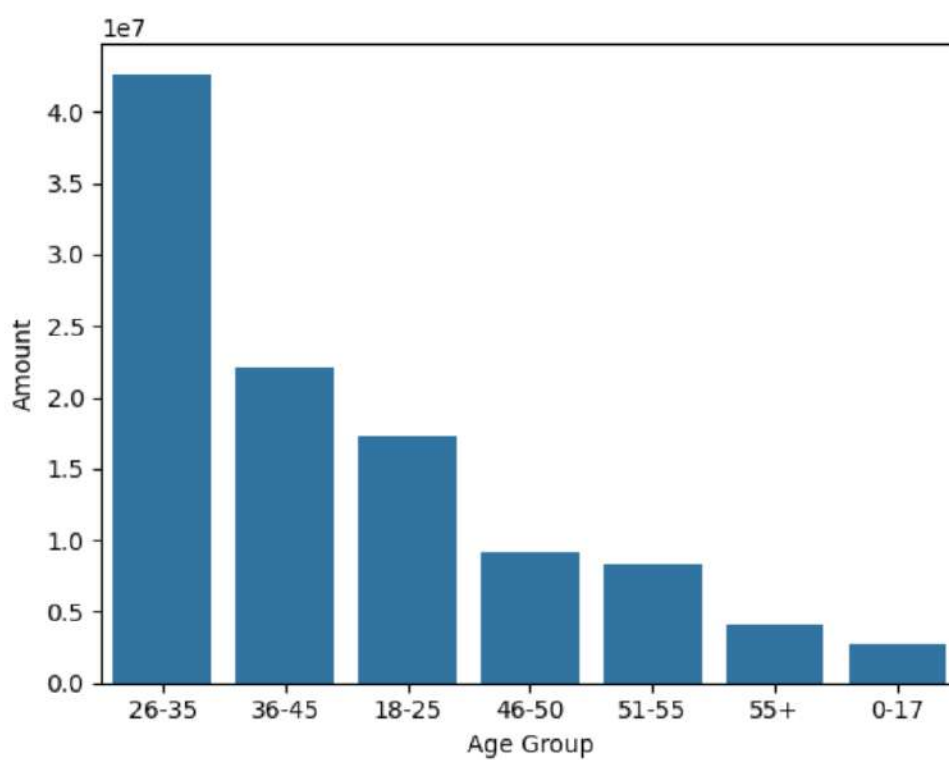
```
[72]: fig,ax = plt.subplots()
      ax.bar(butter_price.key(), butter_price.values());
      ax.set(title = "Butter_price",
            xlabel = "companies",
            ylabel = "prices")
```

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```
[78]: sales_age = df.groupby(['Age Group'], as_index = False)['Amount'].sum().sort_values(by = 'Amount', ascending = False)
      sns.barplot(x = 'Age Group' , y = 'Amount' , data = sales_age)
```

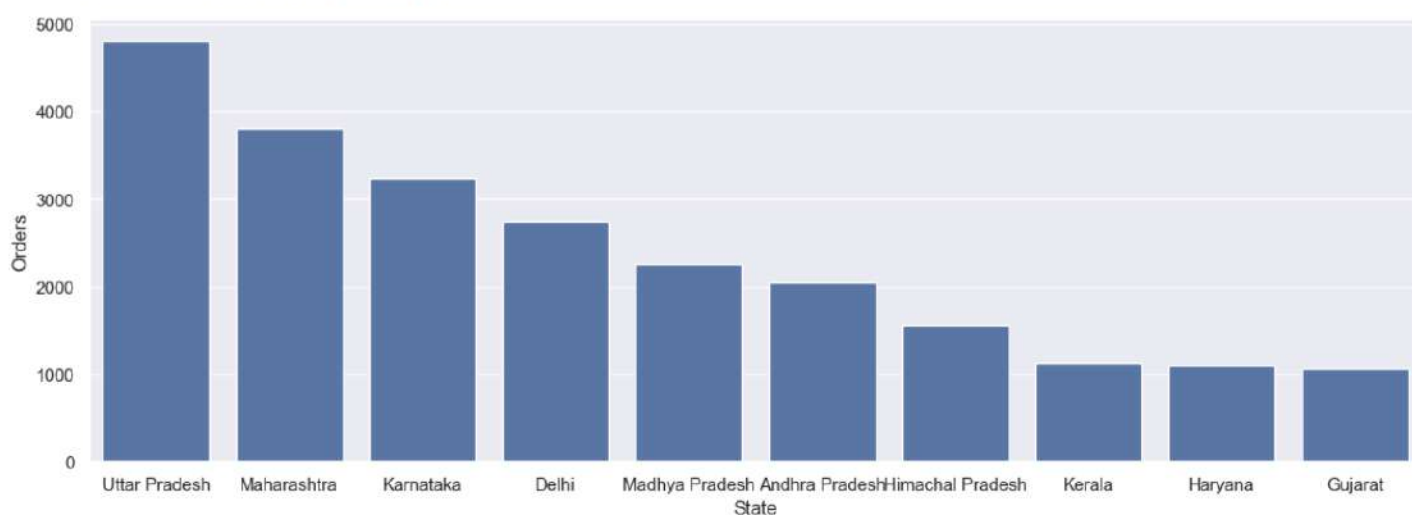
```
[78]: <Axes: xlabel='Age Group', ylabel='Amount'>
```


[78]: <Axes: xlabel='Age Group', ylabel='Amount'>



```
[82]: sales_state = df.groupby(['State'], as_index = False)['Orders'].sum().sort_values(by = 'Orders', ascending = False).  
sns.set(rc={'figure.figsize' : (15,5)})  
sns.barplot(x = 'State', y = 'Orders' , data = sales_state)
```

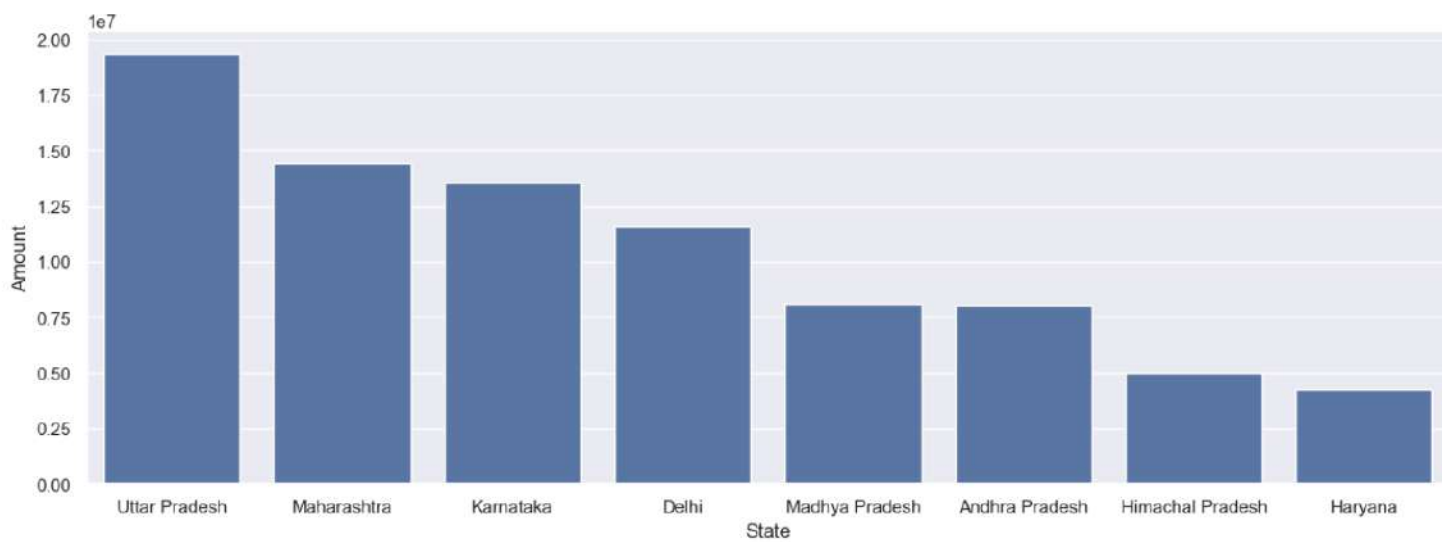
```
[82]: <Axes: xlabel='State', ylabel='Orders'>
```



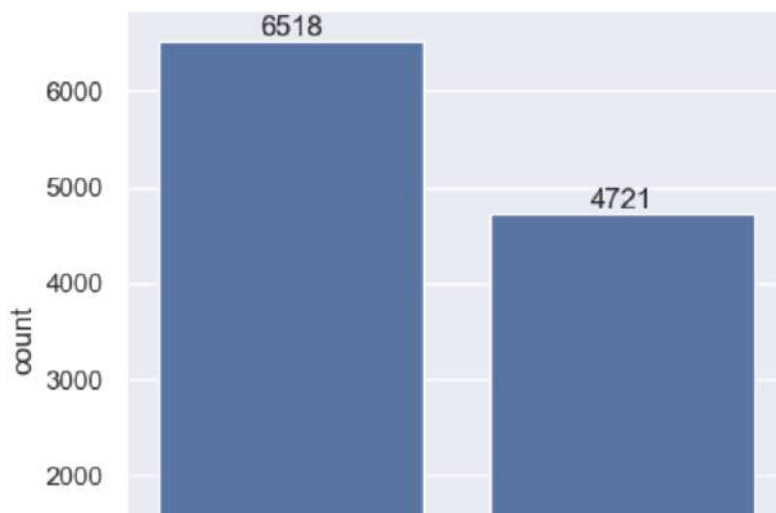
```
[86]: sale_Amount = df.groupby(['State'], as_index = False)['Amount'].sum().sort_values(by = 'Amount', ascending = False).  
sns.set(rc = {'figure.figsize':(15,5)})
```

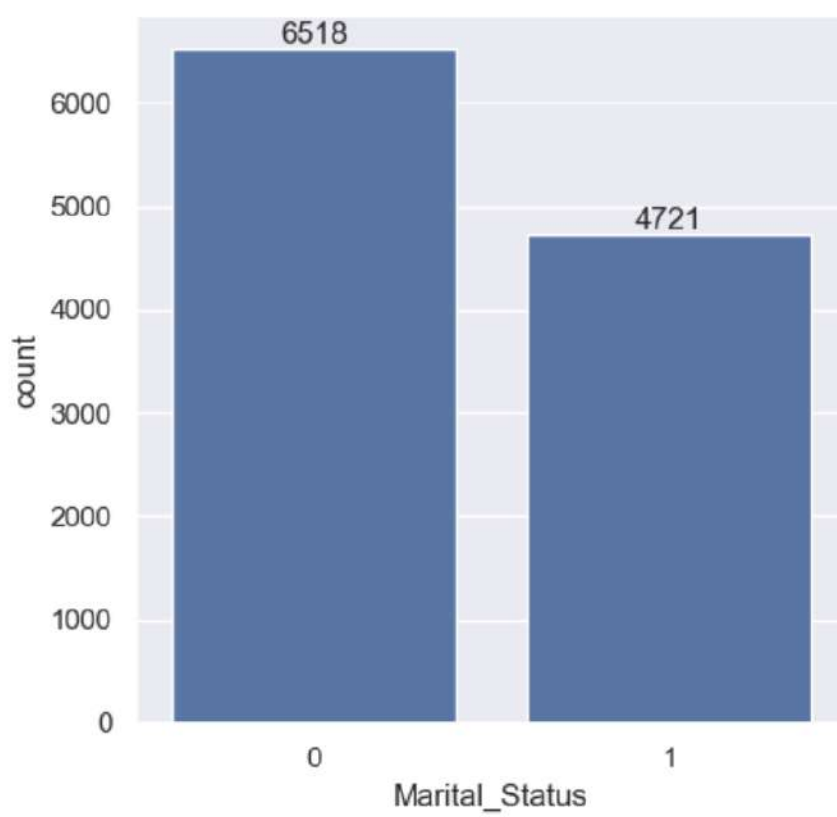
```
[86]: sale_Amount = df.groupby(['State'], as_index = False)['Amount'].sum().sort_values(by = 'Amount', ascending = False)
sns.set(rc = {'figure.figsize':(15,5)})
sns.barplot(data = sale_Amount, x = "State", y = "Amount")
```

```
[86]: <Axes: xlabel='State', ylabel='Amount'>
```



```
[108]: ax = sns.countplot(data = df, x = 'Marital_Status')
sns.set(rc = {'figure.figsize' : (5,3)})
for bars in ax.containers:
    ax.bar_label(bars)
```



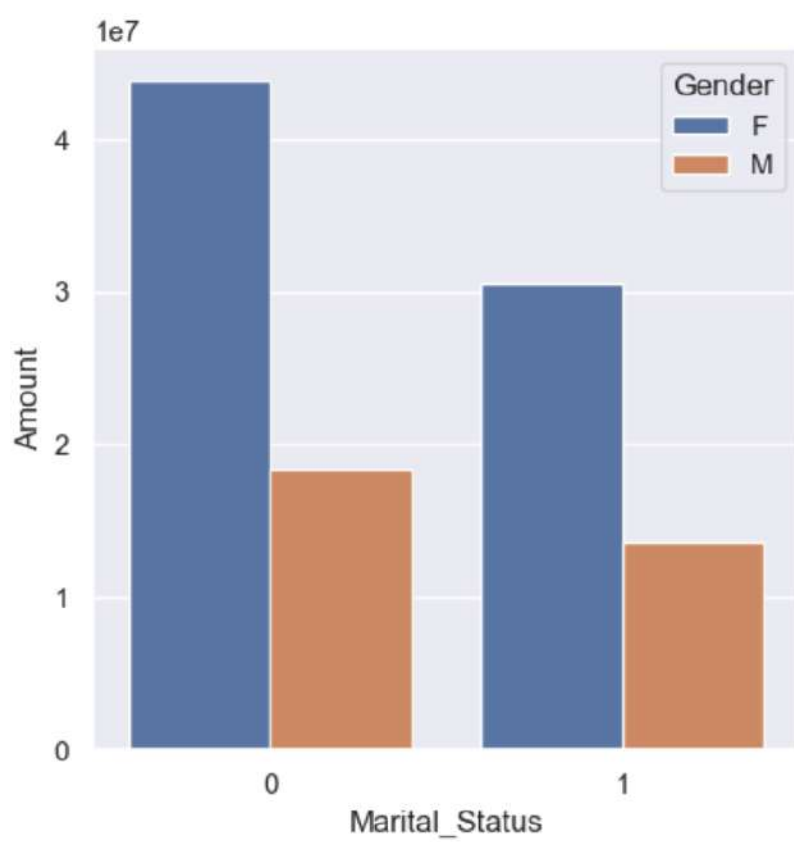


```
[107]: sales_marital_amount = df.groupby(['Marital_Status','Gender'], as_index = False)['Amount'].sum().sort_values(by = 'Amount')

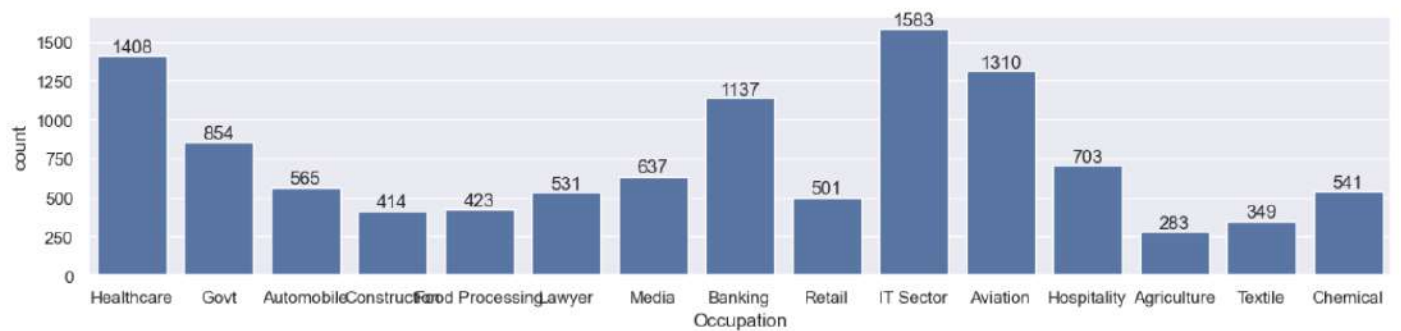
sns.set(rc = {'figure.figsize' : (5,5)})

sales_marital_amount = sns.barplot(data = sales_marital_amount , x = 'Marital_Status' , y= 'Amount', hue = 'Gender')

for bars in ax.containers:
    ax.bar_label(bars)
```



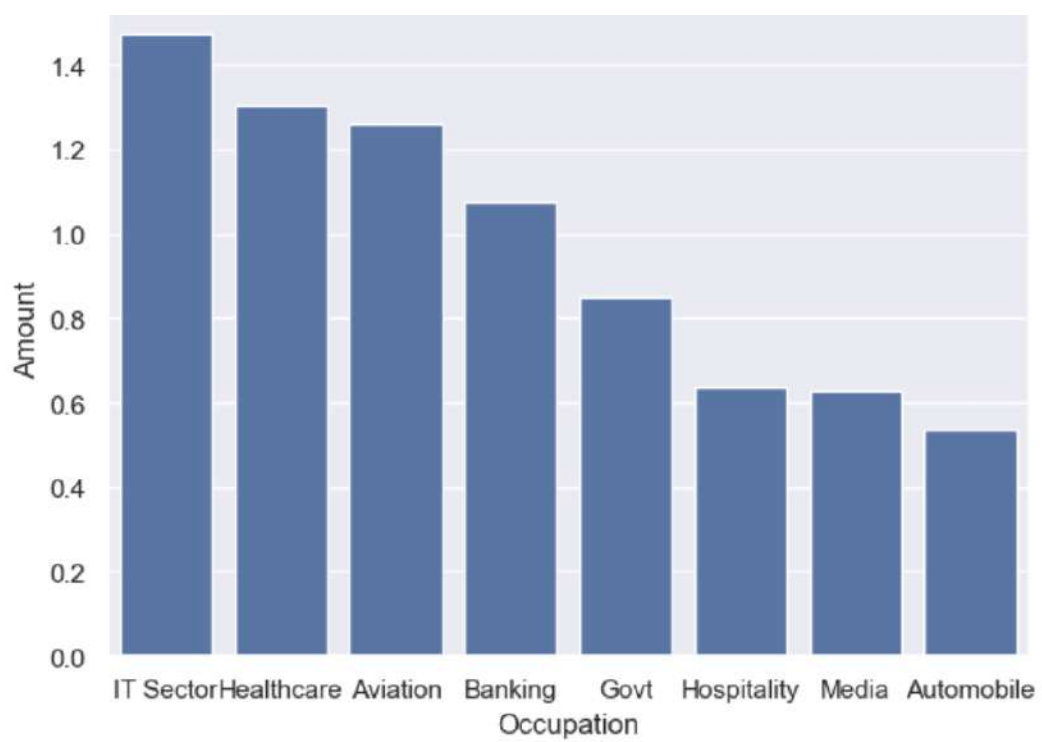
```
[127]: sns.set(rc = {'figure.figsize': (15,3)})
ax = sns.countplot(data = df , x = 'Occupation')
for bars in ax.containers:
    ax.bar_label(bars)
```




```
[111]: df.columns
```

```
[111]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
        'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
        'Orders', 'Amount'],  
        dtype='object')
```

```
[118]: sale_Amount = df.groupby(['Occupation'], as_index = False)['Amount'].sum().sort_values(by = 'Amount', ascending = Fa  
sns.set(rc = {'figure.figsize':(7,5)})  
sns.barplot(data = sale_Amount, x = "Occupation", y = "Amount")
```



```
[124]: sns.set(rc = {'figure.figsize': (20,5)})
ax = sns.countplot(data = df , x = 'Product_Category')
for bars in ax.containers:
    ax.bar_label(bars)
```

